CASE STUDY · SUPPLY CHAIN OPS

D2C Warehouse Pre-Dispatch Teardown & Issue-Triage Playbook

How an order moves from inbound stock to carrier handover, where that flow breaks, and how a first-line analyst fixes or routes each failure within SLA

Charukant · October 2026

Most dispatch delays are two systems disagreeing

In a 37,401-order sample month, the fixes that matter are master data, sync and mapping, not new tools

93.8%

of orders picked up by the carrier within 24 h, against a 95% target

34%

of late orders were blocked at inventory sync, the single largest cause

3 SKUs

fail at the same step on 19–24% of their orders: a wrong bin, a combo mapping, a missing HSN

What this playbook covers

- The 10-step pre-dispatch flow and the system that owns each step
- Where each step breaks, what the floor sees, and who fixes it
- A triage flow, P1–P4 SLA tiers and an escalation ticket template
- An issue log and weekly review that turns repeat failures into fixes

Three systems and two outside parties touch every order

Knowing which system owns a record tells you who can fix it



ERP (SAP)

- Purchase orders
- GRN posting
- SKU, HSN and tax master
- Accounts and stock value



WMS

- Bins and putaway
- Pick and pack scans
- Cycle counts
- Physical stock



OMS (e.g. Unicommerce)

- Channel orders and sync
- Stock allocation
- Picklists, invoices
- Labels and manifests



Carrier

- AWB numbers
- Serviceability by PIN
- Pickup scan
- Tracking



GST portal

- E-invoice IRN (B2B)
- E-way bill
- Part B vehicle details
- Validity and cancellation

An order passes ten steps before a carrier scans it out

Each step leaves one record that proves it finished. That record is the first thing to check when an order is stuck

1

Inbound & GRN

WMS + ERP

Proof: GRN number

2

Putaway

WMS

Proof: Stock at a bin

3

Inventory sync

OMS

Proof: Channel stock = WMS

4

Allocation

OMS

Proof: Order Allocated

5

Picklist

OMS / WMS

Proof: Picklist assigned

6

Pick & pack

WMS

Proof: Package Packed

7

Invoice

OMS → ERP

Proof: Invoice no. (+ IRN)

8

Label & AWB

OMS + carrier

Proof: AWB on package

9

Manifest

OMS

Proof: Manifest closed

10

E-way bill & handover

GST portal + carrier

Proof: Pickup scan

Steps 1–3 make stock sellable; steps 4–10 turn an order into a parcel the carrier accepts

Where the flow breaks: stock and allocation

The analyst's first question is always which two records disagree

Step	Failure	What the floor sees	Likely root cause	Route to
Inbound & GRN	**GRN not posting**	Stock on the dock, not in the system	PO quantity or SKU mismatch; PO closed in ERP	ERP (SAP) support
Inbound & GRN	**Partial or wrong GRN**	Received count differs from invoice	Short or excess supply; QC rejects not recorded	Warehouse lead, procurement
Putaway	**Stock not sellable**	Item on shelf, shows zero available	Putaway not confirmed; bin inactive or quarantined	WMS / OMS support
Inventory sync	**Overselling or false stock-outs**	Orders for items not on the shelf	Sync job failed or delayed; buffer rules wrong	OMS, channel integration
Allocation	**Orders stuck in Unfulfillable**	Picklists missing today's orders	Old reservation blocking stock; facility or combo SKU mapping missing	OMS support
Picklist	**Picklist not generating**	Pickers idle at shift start	Orders on hold; picklist rules exclude a channel	OMS support, ops lead

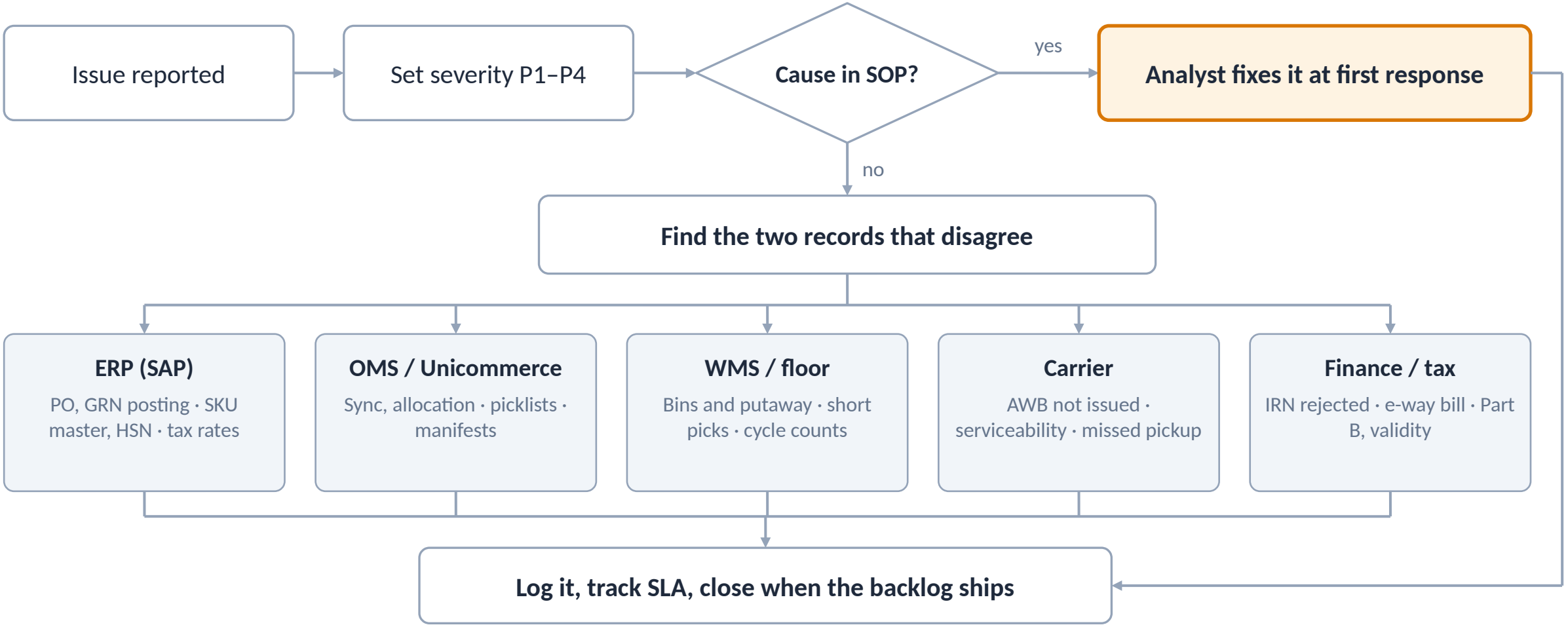
Where the flow breaks: packing to handover

Late-stage failures hold up packed parcels, so they are usually the most urgent

Step	Failure	What the floor sees	Likely root cause	Route to
Pick & pack	**Item not found in bin**	Picker marks a short pick	Physical and system stock differ; wrong putaway	Warehouse lead + WMS
Invoice	**Invoice fails**	Packed, but cannot move to label	Missing HSN or tax rate; GSTIN error; IRN rejected (B2B)	ERP master data, OMS
Label & AWB	**No AWB returned**	Label printer idle; package stuck	Carrier API down; PIN not serviceable; weight rule	Carrier key account
Manifest	**Manifest will not close**	Carrier waiting at the dock	Package in wrong status; cancelled order in manifest	OMS support
E-way bill	**E-way bill not generated**	Truck cannot leave (B2B / transfer)	Value over threshold, no bill; Part B vehicle missing	Finance / tax
Handover	**Shipped in OMS, not picked up**	Customer sees no movement	Pickup scan missed; manifest not shared	Carrier key account

Fix it at first response or route it to the owning team

Every path ends in the same issue log, so self-fixed issues are counted too



Severity follows orders at risk, not bug difficulty

Proposed starting targets, to tune against the brand's own volumes. Workaround first, then root cause

P1

Dispatch stopped for a facility or a top channel

Invoicing fails for every order; main carrier API down

First response	**15 min**
Workaround	**1 hr**
Resolution	**4 hr**

Escalate to: Vendor on call, Supply Chain PM, warehouse head

P2

One channel, carrier or SKU group blocked

Marketplace orders stuck in Unfulfillable after a sync failure

First response	**30 min**
Workaround	**2 hr**
Resolution	**Same day**

Escalate to: Vendor ticket, Supply Chain PM

P3

A few orders or SKUs hit; workaround exists

One SKU missing its HSN code, so its invoices fail

First response	**4 hr**
Workaround	**Same day**
Resolution	**2 days**

Escalate to: Vendor or master-data team

P4

No dispatch impact

Report column wrong; label layout request

First response	**1 day**
Workaround	**None**
Resolution	**5 days**

Escalate to: Backlog, reviewed weekly

Every escalation carries the same seven fields

So SAP, OMS or carrier support can act without calling back

1. **What is broken**
   Scope and since when
2. **Business impact**
   Orders and value at risk, ship-by time
3. **Steps to reproduce**
   Order ID, SKU, screen
4. **Expected vs actual**
   What the record should say
5. **Already checked**
   So support skips the basics
6. **Workaround in place**
   What is shipping meanwhile
7. **Ask and deadline**
   The fix needed, by when

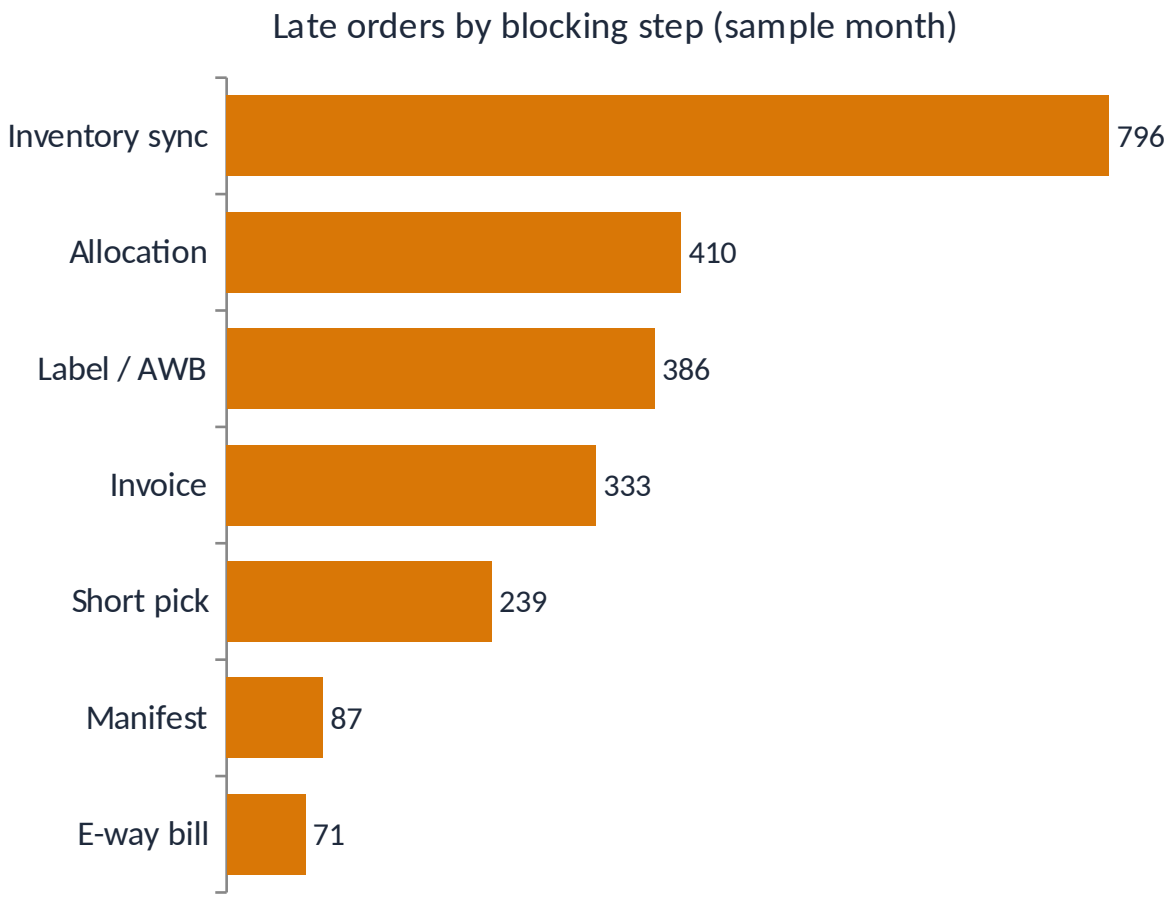
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[P2] Orders stuck in Unfulfillable, WH-01

1  214 orders from one channel stuck since 09:40
2  Approx. Rs 3.1 L miss today's 16:00 ship-by
3  SO-1xxxxx: 40 units in WMS, 0 in OMS
4  Expected Allocated; actual Unfulfillable
5  Putaway confirmed; bin active; last sync 09:15
6  Manual allocation for top 50 orders by ship-by
7  Restore sync, re-run allocation by 13:00

Evidence: screenshots, order IDs (CSV), API error
Owner: analyst on shift · raised 10:05
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Log every issue, including the ones analysts fix themselves

Repeat failures only show up when the small ones are counted too



Weekly 30-minute review

1. SLA adherence by severity and support team, against last week
2. Top three failure types by count and by orders delayed
3. Anything seen 3+ times becomes a fix request with an owner and a date
4. Each new root cause becomes an SOP entry, so it is fixed at first response next time

Log fields: time raised · step · failure type · severity and orders at risk · routed to · workaround · response and resolution time · SLA met · root cause and fix

Two GST rules can stop a B2B or transfer load from leaving

Single B2C parcels rarely need either; replenishment, B2B orders and inter-warehouse transfers do

E-invoice (IRN)

Mandatory on B2B invoices when annual turnover is above Rs 5 Cr

Check: IRN on every B2B invoice before label and manifest

IRN reporting window

Turnover of Rs 10 Cr or more: report invoices within 30 days (since 1 Apr 2025)

Check: no old invoices left unreported until dispatch

E-way bill threshold

Inter-state consignments above Rs 50,000; intra-state limits vary by state

Check: consignment value and state limit before loading

E-way bill validity

1 day per 200 km from Part B entry; documents must be under 180 days old; cancel within 24 h

Check: generate close to dispatch; reissue old invoices

Sources: ClearTax, e-way bill rules (cleartax.in); CalcGuru, e-invoicing thresholds (calcguru.in). Checked October 2026.

A KPI dashboard makes the patterns visible every morning

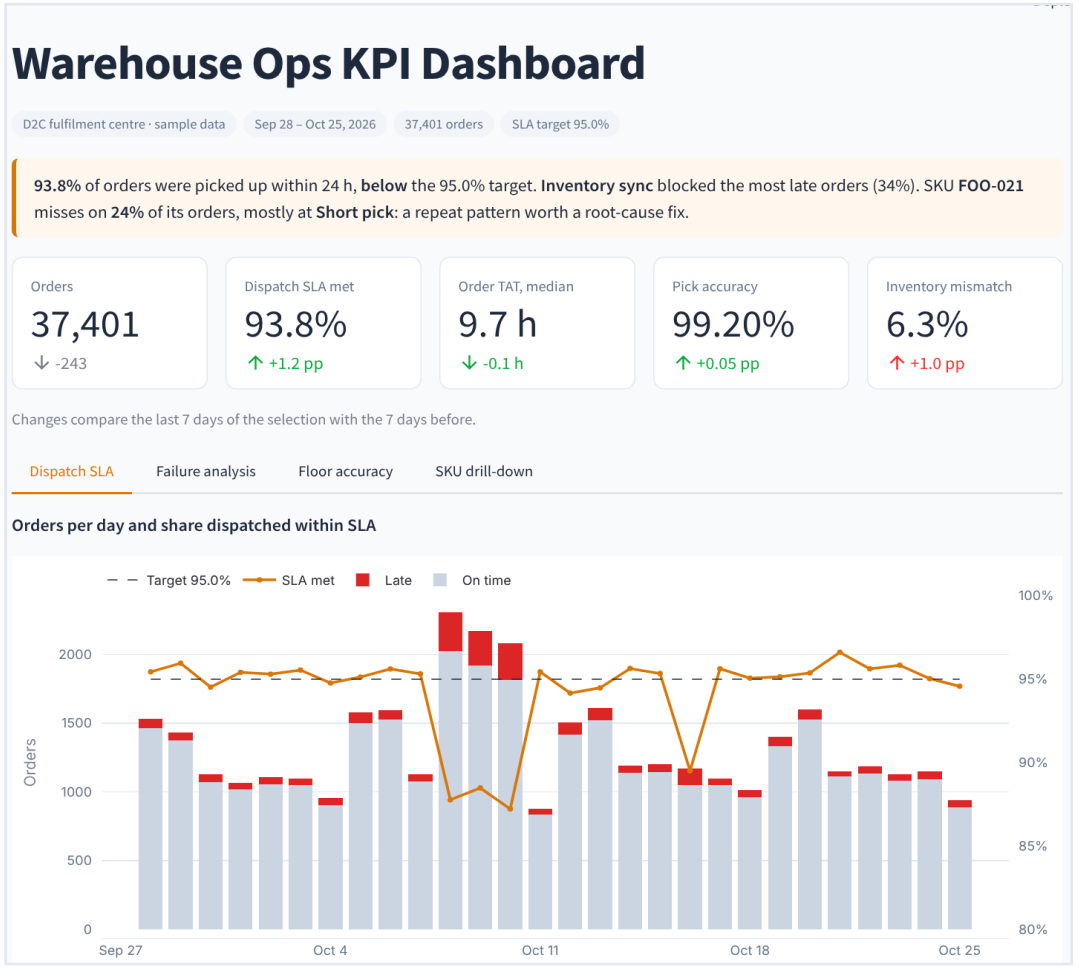
Built in Streamlit on a 37,401-order sample month: SLA, TAT, pick accuracy, stock mismatch, with drill-downs

93.8% dispatch SLA met; festive-sale days and one carrier label outage drove the worst days

34% of late orders blocked at inventory sync: fix the sync job and buffer rules first

9.0% evening late rate on Flipkart, the worst channel-shift pair

3 SKUs late on 19–24% of orders, each at one step: raise three fix requests



How I would run this in the first 30 days

WEEK 1

Walk the floor

Map the real pre-dispatch flow at each site; list every system hand-off and who owns it

Outcome **Flow map and owner list**

WEEK 2

Start the log

Issue log live for every analyst; P1–P4 tiers and the escalation template agreed with SAP, OMS and UC support

Outcome **Every issue logged**

WEEK 3

Fix the repeats

Baseline SLA adherence; take the top three repeat failures to root cause and raise fix requests

Outcome **3 fix requests raised**

WEEK 4

Set the rhythm

Weekly review with the Supply Chain PM; KPI dashboard shared; SOPs updated from every new root cause

Outcome **Weekly KPI review running**